

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

The web server is experiencing a TCP SYN Flood Attack, which is a type of Denial of Service (DoS) attack. The attacker is overwhelming the server with connection requests, preventing legitimate users from accessing the website.

The logs show that:

- A large number of TCP SYN packets are being sent from an unfamiliar IP address.
- The server is receiving more connection requests than it can handle.
- Legitimate users are experiencing connection timeout errors when trying to access the website.
- The web server is becoming overwhelmed and unable to respond normally.

This event could be:

A TCP SYN Flood Attack (Denial of Service attack). The attacker is exploiting the TCP three-way handshake process by sending a large volume of SYN requests without completing the connection, causing the server to exhaust its resources and become unavailable to legitimate users.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a SYN packet to the server requesting a connection.
2. The server responds with a SYN/ACK packet acknowledging the request and indicating it is ready to establish the connection.
3. The client sends an ACK packet back to the server, completing the handshake and establishing the connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at

once:

When a malicious actor sends a large number of SYN packets, the server attempts to respond to each request with a SYN/ACK packet and reserves resources for every pending connection. Because the attacker does not complete the handshake, many half-open connections remain active. This consumes server resources and prevents legitimate users from establishing connections.

Explain what the logs indicate and how that affects the server:

The logs indicate an unusually high number of TCP SYN requests originating from an unfamiliar IP address. This suggests a SYN flood attack is occurring. As the server attempts to process these requests, its resources become exhausted, resulting in slow performance, connection timeouts, and the inability of legitimate users to access the company's website.